ZS-S16

The Z-Spin-Mediated μ–τ Breaking Sum Rule

A Rephasing-Invariant PMNS Spurion Derivation of a Geometric First-Order Leptonic CP Phase from the Primitive-Cell Wilson Holonomy, with the arctan(A) Chart Demotion and a Contragredient Flavor-Orientation Theorem

Author: **Kenny Kang**

Affiliation: Z-Spin Cosmology Collaboration

Date: June 2026

Theme / Paper Code: **Standard Model Completion [ZS-S] | Paper 16**

Version: **v1.2 (June 2026)**

Verification: **32/32 PASS** | **Zero Free Parameters** | (A, Q, dim Z) = (35/437, 11, 2) LOCKED

§0. Abstract

The leptonic Dirac CP phase δ_{CP} is the least-determined parameter of the PMNS matrix. The Z-Spin corpus has carried a candidate prediction $\delta_{CP} = \pm\pi/2 + \arctan(\boldsymbol{A})$ (ZS-Q5 §4, ZS-S9 §5.1), in which a bare maximal phase fixed by μ–τ reflection symmetry is offset by arctan(35/437) = 4.579°. We show that this surface form is **weakly grounded**: the arctan offset has no action-level derivation, it is observationally indistinguishable from a linear offset, and selecting it specifically is a numerology hazard.

We reconstruct the prediction on a far stronger footing. The physical first-order content is not arctan($\boldsymbol{A}$) but the locked geometric impedance $\boldsymbol{A}$ = 35/437 itself, promoted to the primitive-cell CP phase kick:

$$\delta_{CP} = \delta_{\mu\tau} + s_Z \cdot \boldsymbol{A} + O(\boldsymbol{A}^3)$$

where $\delta_{\mu\tau} = \pm\pi/2$ is imported standard physics (μ–τ reflection), $s_Z = \pm1$ is the Z-boundary branch, and the kick magnitude is the **primitive-cell Wilson holonomy** $\oint \omega = \boldsymbol{A}$ established as DERIVED via Gauss–Bonnet on the polyhedral defect manifold (ZS-F0/F-rotor §5.8, cross-verified in ZS-U5, ZS-A6, ZS-M3). The kick is rephasing-invariant because a Wilson loop is gauge-invariant. We prove (Theorem S16.2) that $\arctan(\boldsymbol{A}) = \boldsymbol{A} - \boldsymbol{A}^3/3 + O(\boldsymbol{A}^5)$, so the two forms differ by only 0.01° at 35/437 and the arctan form is a resummation chart (the principal value of a Cayley impedance), not an independent physical prediction. The Jarlskog invariant inherits a parameter-free suppression $J/J_{bare} = \cos \boldsymbol{A} = 1 - \boldsymbol{A}^2/2 + O(\boldsymbol{A}^4)$ = 0.99679 (0.321% suppression). The negative branch $s_Z = -1$ is fixed (DERIVED-CONDITIONAL) by the spin-½ boundary lift $\chi_Z = -1$ (ZS-Q14) composed with the contragredient channel $V_{ZY} = (V_{XZ})^*$, selecting $\delta_{CP} = -\pi/2 - \boldsymbol{A}$ = 265.41° (inverted ordering).

Against NuFIT 6.0 (Sept 2024) the prediction sits within ≈0.7σ of the inverted-ordering best fit (285°) but is in tension with the now-preferred normal ordering ($\delta_{CP} \approx$ 177°, near CP conservation); we therefore register the sharp value as an **IO-conditional kill gate**, decisively testable by DUNE and Hyper-Kamiokande in the 2030s. We further register two well-posed open problems of external interest: (i) the exact arctan form survives only if an open-channel retarded Feshbach self-energy yields the ratio $\Gamma_Z/(2\Delta_Z) = \boldsymbol{A}$ [OPEN]; and (ii) a Lindblad dissipative reading of the same Z-mediator (coefficient 2$\boldsymbol{A}$/Q, ZS-Q15) predicts a distinct decoherence-driven J suppression that does not shift δ_{CP} — a fork the data can separate in principle. Zero free parameters beyond the locked triple.

New in v1.1. We upgrade the sum rule from an embedding postulate to a **rephasing-invariant mass-matrix derivation** (Lemma S16.5): the Z-Spin breaking enters as a real, off-diagonal, μ–τ-seam-odd, CP-odd spurion $\Delta_Z(\boldsymbol{A}) = i\cdot s_Z\cdot\boldsymbol{A}\cdot B_{\mu\tau}$ whose off-diagonal support makes it orthogonal to the entire charged-lepton rephasing algebra — so it cannot be absorbed by any field redefinition (it is not a diagonal/Majorana rephasing) and sources a genuine, rephasing-invariant Dirac phase. The Jarlskog commutator invariant then scales exactly as $\mathrm{Im}[(H_\nu)_{e\mu}(H_\nu)_{\mu\tau}(H_\nu)_{\tau e}]$ = (bare) $\cos\boldsymbol{A} + O(\boldsymbol{A}^3)$, closing the registered corpus open problem ZS-S5 OP-(ii) ("mapping from $\boldsymbol{A}$ to PMNS matrix elements"). We also close the full-angle/half-angle ambiguity: the PMNS Dirac phase is sourced by the rephasing-invariant **relative contragredient transfer phase** $\arg(V_{XZ}) - \arg(V_{ZY}) = 2\cdot(\boldsymbol{A}/2) = \boldsymbol{A}$ (the holonomy), not by either half-path amplitude phase $\boldsymbol{A}/2$ separately — so $\boldsymbol{A}$ is selected and $\boldsymbol{A}/2$ is a closed rebuttal (F-S16.5 PASS). Branch selection is separated into a Contragredient Flavor-Orientation Theorem (S16.6): the relative orientation $s_Z = -1$ is DERIVED-CONDITIONAL, while the absolute μ-vs-τ label remains the lone residual OPEN (ZS-S10 Gap G1). v1.2 then positions that lone conditional precisely: the embedding Z = ∂X is identified with the **single irreducible postulate of the entire Z-Spin corpus** (the ZS-Q12V §13–§16 Bedrock cascade Z = ∂X ⇒ 4π ⇒ i-tetration ⇒ z*; ZS-Q16 v2.5 §22 confirms it cannot be reduced further), so the sum rule is as closed as any Z-Spin result and its unconditional DERIVED status is pending only the corpus-wide closure of Z = ∂X (O-Q16.12), not anything internal to ZS-S16.

Keywords: leptonic CP violation, PMNS, μ–τ reflection symmetry, Jarlskog invariant, rephasing invariance, flavor spurion, Wilson holonomy, geometric phase, contragredient transfer, Cayley chart, retarded Feshbach self-energy, Z-Spin Cosmology, zero free parameters.

§0.1 Epistemic Status Legend

STATUS	Definition
PROVEN	Mathematical theorem; follows from standard mathematics or corpus definitions alone.
LOCKED	Core constant fixed upstream; no downstream paper may modify.
DERIVED	Follows from the Z-Spin action plus PROVEN/LOCKED inputs; zero free parameters.
DERIVED-CONDITIONAL	Derived under one stated upstream postulate (here: Z = ∂X, a physical spinor boundary).
VERIFIED	Numerically confirmed against data or independent computation.
TESTABLE	Well-posed prediction with a pass/fail experimental gate.
IMPORTED / STANDARD	Established external (non-Z-Spin) result, stated to locate the construction.
HYPOTHESIS-weak	Motivated but derivation absent and anti-numerology not passable as posed.
NON-CLAIM	Explicitly not asserted; a chart/coordinate artifact, documented to prevent overclaim.
COROLLARY	Immediate consequence of a stated Lemma/Theorem; inherits its status.
OPEN	Well-posed unresolved problem.

§1. Introduction

1.1 The leptonic CP phase and μ–τ reflection

The Pontecorvo–Maki–Nakagawa–Sakata (PMNS) matrix carries one Dirac CP phase δ_{CP}. Among the six standard oscillation parameters it is the least constrained: global fits determine it only at the tens-of-degrees level and its value depends on the assumed neutrino mass ordering [1]. A long-standing theoretical anchor is μ–τ reflection symmetry [4]: invariance of the neutrino mass matrix under the combined μ↔τ exchange and complex conjugation forces $|U_{\mu i}| = |U_{\tau i}|$, hence $\theta_{23} = \pi/4$ and **maximal CP violation $\delta_{CP} = \pm\pi/2$**. We import this bare value as STANDARD physics and write $\delta_{\mu\tau} = \pm\pi/2$.

1.2 The arctan(A) proposal and why it is weak

ZS-Q5 §4 and ZS-S9 §5.1 propose that the Z-mediated transit of a fermion through the planar Z-sector imposes a geometric impedance phase $\delta_\varphi = \boldsymbol{A}$, which, under a "gauge-manifold projection," enters as an offset arctan($\boldsymbol{A}$) = 4.579°, giving $\delta_{CP} = \pm\pi/2 + \arctan(\boldsymbol{A})$. Four independent objections weaken this surface form: (i) the corpus itself registers the arctan correction as lacking an action-level QFT derivation; (ii) the half-angle holonomy actually present in ZS-F4 §7B is $e^{i\theta/2}$ with $\theta = \pi(1-\varepsilon)$ position-dependent — not arctan($\boldsymbol{A}$); (iii) arctan($\boldsymbol{A}$) differs from $\boldsymbol{A}$ by 0.2%, far below any foreseeable experimental resolution, so the two are observationally identical; and (iv) within the family $\{ \boldsymbol{A}, \arctan \boldsymbol{A}, \boldsymbol{A} - \boldsymbol{A}^3/3, \ldots \}$ every member fits equally, so isolating arctan is a numerology hazard in the precise sense of the corpus anti-numerology methodology (ZS-S9 App. B).

1.3 This paper: promote A, demote arctan

Our thesis is a single sentence: **Z-Spin predicts a one-cell CP phase kick *A*, not necessarily arctan(*A*).** The reasons are structural. First, $\boldsymbol{A}$ = 35/437 is LOCKED (ZS-F2) and, crucially, equals the **primitive-cell Wilson holonomy** $\oint \omega = \boldsymbol{A}$ — a DERIVED quantity, not a fitted one (§2.1). Second, a Wilson loop is gauge/rephasing-invariant by construction, so a CP phase identified with it is automatically physical (not removable by field redefinition). Third, arctan($\boldsymbol{A}$) is then exactly the Taylor/Cayley resummation of this leading kick, $\boldsymbol{A} - \boldsymbol{A}^3/3 + \ldots$, demoted from a physical claim to a coordinate convention. This single move dissolves objections (i)–(iv) at once and converts the open question from "why arctan?" to the sharper "what fixes the coefficient and sign of the $\boldsymbol{A}$-kick?", which we answer to DERIVED-CONDITIONAL level.

1.4 Scope: claims and non-claims

DERIVED / DERIVED-CONDITIONAL targets: the sum rule $\delta_{CP} = \pm\pi/2 + s_Z \boldsymbol{A}$ (Thm S16.1); the arctan chart equivalence (Thm S16.2, PROVEN); the Jarlskog suppression cos $\boldsymbol{A}$ (Thm S16.3); the branch sign $s_Z = -1$ (Thm S16.4).

NON-CLAIMS: that arctan($\boldsymbol{A}$) is the exact physical phase (it is a chart); that δ_{CP} = 265.41° is a currently-closed prediction (it is IO-conditional); that the leptonic Jarlskog is measured (it is not). New free parameters: **zero** beyond ($\boldsymbol{A}$, Q, dim Z) = (35/437, 11, 2).

§2. Locked Inputs and the Primitive-Cell Wilson Holonomy

Table 1. Locked inputs for the μ–τ breaking sum rule.

Quantity	Value	Source	Status
***A* (impedance)**	35/437 = 0.0800915	ZS-F2	**LOCKED**
(Z, X, Y); Q	(2, 3, 6); 11	ZS-F5	**PROVEN**
$\oint \omega = \boldsymbol{A}$	35/437	ZS-F0/F-rotor §5.8	**DERIVED**
V_{XZ} phase	$e^{i\theta/2}$	ZS-F4 §7, §7B	**DERIVED-COND**

Quantity	Value	Source	Status
$V_{ZY} = (V_{XZ})^*$	CPT conjugate	ZS-A7 Cor I, ZS-S9	**DERIVED**
$j = 1/2$; $D(2\pi)=-I$	dim(Inv)=2=Z	ZS-M3 Thm 5.1	**PROVEN**
$\chi_Z = -1$	half-angle lift	ZS-Q14 Lem 5.1	**DERIVED-COND**
$\delta_{\mu\tau}$	$\pm\pi/2$	Harrison–Scott [4]	**IMPORTED**

2.1 The Wilson holonomy $\oint \omega = A$ (the anchor)

The geometric impedance $\boldsymbol{A}$ acts as a connection 1-form ω on the $O_h \times I_h$ polyhedral defect manifold of ZS-F2. Parallel transport around one primitive cell accumulates the Wilson-loop holonomy

$$\delta\varphi_{cell} = \oint_{primitive\ cycle} \omega = \boldsymbol{A} = 35/437,$$

established as DERIVED in ZS-F0/F-rotor §5.8 by the Gauss–Bonnet theorem on the polyhedral defect manifold (ZS-F4 §3 PROVEN). The same holonomy fixes the first-rotation rate $\theta(0)\cdot\tau_P = \boldsymbol{A}$ and, on cosmological parallel transport, the macroscopic Hubble ratio $H_0^{local}/H_0^{CMB} = \exp(\oint \omega) = \exp(\boldsymbol{A}) = 1.0834$ (matched to SH0ES at 0.06σ). The Planck-scale identity $\delta\varphi_{cell} = \boldsymbol{A}$ is independently cross-verified in ZS-U5 Lemma 8.1, ZS-A6 §5.2, and ZS-M3 §5. This is the single geometric quantity our sum rule rests on, and it is DERIVED rather than asserted — the decisive contrast with arctan($\boldsymbol{A}$).

2.2 The half-angle spinor channel

The Z-mediated transfer carries a half-angle phase: $V_{XZ}(r) = \sqrt{\boldsymbol{A}} \cdot \varepsilon/\sqrt{(1+\boldsymbol{A}\varepsilon^2)} \cdot e^{i\theta/2}$ with $\theta(r) = \pi(1-\varepsilon)$, and $V_{ZY} = (V_{XZ})^*$ the contragredient conjugate (ZS-F4 §7B, DERIVED-CONDITIONAL). The half-angle is the spinor signature $D(2\pi) = -I$ of the unique j = 1/2 intertwiner forced by dim(Inv) = 2 = dim(Z) (ZS-M3 Thm 5.1, PROVEN). The product $V_{ZY}\cdot V_{XZ}$ is real (Im = 0, the Dimension-Ratio constraint), but the **relative phase** $\arg(V_{XZ}) - \arg(V_{ZY}) = \theta$ is the gauge-invariant, physically observable quantity (ZS-S15 §6). Over one primitive cell this relative phase is the holonomy $\boldsymbol{A}$ of §2.1.

§3. The μ–τ Breaking Sum Rule

3.1 Statement

Theorem S16.1 (Wilson-Holonomy CP Seed, DERIVED-CONDITIONAL). Embedding the primitive-cell Z-mediation holonomy into the μ–τ-exchanging Z_2 seam channel of the lepton seesaw injects a rephasing-invariant, CP-odd phase whose leading order is the holonomy itself:

$$\delta_{CP} = \delta_{\mu\tau} + s_Z \cdot \boldsymbol{A} + O(\boldsymbol{A}^3), \quad \delta_{\mu\tau} = \pm\pi/2, \quad s_Z \in \{-1, +1\}.$$

3.2 Lemma S16.5 — Rephasing-Invariant PMNS Spurion Lemma (the invariant backbone)

Lemma S16.5 (DERIVED-CONDITIONAL; scaling PROVEN). Let the charged-lepton mass basis be fixed (ZS-M10 generation assignment). The Z-Spin breaking of the μ–τ-protecting Z_2 seam (ZS-F5 κ = 4 witness; ZS-S5 App. B bounded spurion, ε = O($\boldsymbol{A}$)) enters the light-neutrino mass matrix as

$$M_\nu(\boldsymbol{A}) = M_\nu^{(\mu\tau)} + \Delta_Z(\boldsymbol{A}), \quad \Delta_Z(\boldsymbol{A}) = i\, s_Z \boldsymbol{A}\, B_{\mu\tau} + O(\boldsymbol{A}^2),$$

where $B_{\mu\tau}$ is real, Hermitian, **off-diagonal** in the (e, μ, τ) basis, and μ–τ-seam-odd ($G B_{\mu\tau} G = -B_{\mu\tau}$, G the ZS-F5 seam involution). Then the rephasing-invariant Jarlskog commutator scales as

$$Im[\, (H_\nu)_{e\mu}(H_\nu)_{\mu\tau}(H_\nu)_{\tau e}\,] = Im[\, (H_\nu^{(0)})_{e\mu}(H_\nu^{(0)})_{\mu\tau}(H_\nu^{(0)})_{\tau e}\,] \cos \boldsymbol{A} + O(\boldsymbol{A}^3), \quad H_\nu = M_\nu M_\nu^\dagger.$$

Proof (two parts). (P1) Non-absorbability [PROVEN]. The charged-lepton rephasing group is the diagonal subgroup; its Lie algebra is spanned by diag(1,0,0), diag(0,1,0), diag(0,0,1). For any diagonal D, the Hilbert–Schmidt pairing $\langle B_{\mu\tau}, D\rangle = \mathrm{Tr}(B_{\mu\tau} D) = \Sigma_i (B_{\mu\tau})_{ii} D_{ii} = 0$ because $B_{\mu\tau}$ is off-diagonal ($(B_{\mu\tau})_{ii} = 0$). Hence Δ_Z is orthogonal to the full rephasing algebra and cannot be removed by any charged-lepton field redefinition. This is the decisive point: a *diagonal* spurion (even a seam-odd one such as diag(0,1,−1)) would be a rephasing and would be absorbed into the unphysical PMNS phases (Majorana/field convention); the off-diagonal Δ_Z is not, so it sources a physical rephasing-invariant Dirac phase. (P1)

(P2) cos *A* scaling [PROVEN given the spurion type]. The factor i renders Δ_Z CP-odd (under $M \to M^*$, $\Delta_Z \to -\Delta_Z$); together with G-oddness it is the CP×(μτ) order parameter. At the μ–τ fixed point the CP-odd part of the breaking is conjugate to the δ direction, while the CP-even part would shift θ_{23}; therefore at $O(\boldsymbol{A})$ the spurion rotates the phase, $\delta \to \delta_{\mu\tau} + s_Z\boldsymbol{A}$, and leaves $\theta_{23} = \pi/4$ unchanged to $O(\boldsymbol{A})$. Since the Jarlskog prefactor $c_{23}s_{23} = ½ \sin 2\theta_{23}$ is **stationary** at $\theta_{23} = \pi/4$ (a maximum, so its variation is $O(\boldsymbol{A}^4)$), the invariant scales purely through the phase: $\mathrm{Im}[\ldots]/\mathrm{Im}[\ldots]^{(0)} = \sin(\pm\pi/2 + s_Z\boldsymbol{A})/\sin(\pm\pi/2) = \cos \boldsymbol{A}$. The $O(\boldsymbol{A}^3)$ remainder comes from the $O(\boldsymbol{A}^2)$ piece of Δ_Z inducing an $O(\boldsymbol{A}^2)$ phase shift: $\cos(\boldsymbol{A} + c\boldsymbol{A}^2) = \cos \boldsymbol{A} - c\boldsymbol{A}^3 + O(\boldsymbol{A}^4)$. (P2)

Status. The scaling (P1)–(P2) is PROVEN given the spurion symmetry type. The spurion type and magnitude *A* are DERIVED-CONDITIONAL (CP-oddness from the contragredient conjugation $V_{ZY} = (V_{XZ})^*$ of ZS-F4 §7B; seam-oddness from ZS-F5; magnitude *A* from the Wilson holonomy of §2.1; conditional on the embedding Z = ∂X). Lemma S16.5 thereby closes ZS-S5 Open Problem (ii) — the mapping from *A* to PMNS matrix elements — and removes the "physical-or-rephasing?" objection that left S16.1 at the level of an assertion in v1.0.

3.3 Derivation chain

(D1) μ–τ reflection (STANDARD [4]) fixes the bare matrix to $\theta_{23} = \pi/4$ and $\delta_{CP} = \pm\pi/2$; the μ↔τ exchange is implemented by the Z_2 seam involution $W^2 = I$ (ZS-F5).

(D2) Breaking the exact Z_2 reflection is controlled by the impedance *A*: the leading non-vanishing splitting is $O(\boldsymbol{A})$ (ZS-S5 §3.2; "$\boldsymbol{A} \neq 0 \Rightarrow J_{CP} \neq 0$", ZS-S5 §8). Thus the symmetry-breaking spurion has magnitude set by *A*, not by a free coupling.

(D3) The breaking enters as the relative phase of the CPT-conjugate transfer amplitudes V_{XZ} and $V_{ZY} = (V_{XZ})^*$ (§2.2). Because these are exact conjugates, their relative phase is T-odd, hence CP-odd: it cannot be a CP-even mixing-angle shift.

(D4) Over one primitive cell the relative phase equals the Wilson holonomy $\oint \omega = \boldsymbol{A}$ (§2.1, DERIVED). A Wilson loop is gauge-invariant, so the induced phase is rephasing-invariant. Lemma S16.5 (P1) makes this precise: the breaking is the off-diagonal spurion Δ_Z, orthogonal to the rephasing algebra, hence a genuine δ_{CP} contribution — not a field-redefinition artifact and not a diagonal/Majorana rephasing.

(D5) Magnitude. Because the kick is exactly one cell's holonomy, the coefficient is unity (c = 1): the kick is *A*, not c·*A*. This is the structural advantage over arctan: the magnitude is **fixed by a closed-form holonomy**, with no continuous tuning.

Status. DERIVED-CONDITIONAL, now on the invariant footing of Lemma S16.5 (the physicality is PROVEN; the conditional is the embedding Z = ∂X, i.e., that the lepton μ–τ channel carries the primitive-cell holonomy, inherited from ZS-Q12).

3.4 The full-angle vs half-angle question, resolved by rephasing invariance (Patch 2)

v1.0 left the magnitude convention (kick = ***A*** vs ***A***/2) as an open gate (F-S16.5). It is now closed. The contragredient amplitudes carry equal and opposite half-path phases,

$$arg(V_{XZ}) = +\boldsymbol{A}/2, \quad arg(V_{ZY}) = -\boldsymbol{A}/2,$$

so the **relative** contragredient transfer phase is

$$arg(V_{XZ}) - arg(V_{ZY}) = arg(V_{XZ}^2) = 2\cdot(\boldsymbol{A}/2) = \boldsymbol{A} = \oint \omega.$$

The decisive proposition: **the PMNS Dirac phase is sourced by the rephasing-invariant relative contragredient transfer phase arg(V_{XZ}) − arg(V_{ZY}), not by either half-path amplitude phase separately.** The relative phase is rephasing-invariant — it is the argument of the ratio $V_{XZ}/V_{ZY} = V_{XZ}/V_{XZ}^* = V_{XZ}^2/|V_{XZ}|^2$, and it equals the gauge-invariant holonomy ***A***. By contrast the single half-path phase ***A***/2 is **not** rephasing-invariant: a redefinition of the Z-field phase, $V_{XZ} \to e^{i\chi}V_{XZ}$, $V_{ZY} \to e^{-i\chi}V_{ZY}$, shifts each half-path phase but leaves the relative phase (and ***A***) fixed. Since δ_{CP} is rephasing-invariant (Lemma S16.5), it reads ***A***, not ***A***/2.

Conclusion: the kick is the full holonomy ***A*** = 4.589°; the half-angle reading ***A***/2 = 2.294° is the non-invariant single-path amplitude phase and is therefore a **closed rebuttal** (F-S16.5 PASS), not an open alternative.

3.5 Bedrock positioning: the conditional Z = ∂X is the single irreducible postulate of Z-Spin

A natural question is whether the lone conditional of Theorem S16.1 and Lemma S16.5 — the embedding Z = ∂X — should be discharged within this paper. It should not, and it cannot be reduced to anything more primitive. We position it precisely, because doing so **raises** the result's standing rather than weakening it.

ZS-Q12V §13–§16 (Bedrock Reduction) establishes, as a chain of PROVEN theorems, that the entire Z-Spin framework rests on this single postulate:

$$Z = \partial X \;\Rightarrow\; \dim Z = 2 \;\Rightarrow\; SO(3) \text{ frame} \;\Rightarrow\; \pi_1 = \mathbb{Z}/2 \;\Rightarrow\; 4\pi \;\Rightarrow\; b = i \;\Rightarrow\; \text{i-tetration} \;\Rightarrow\; \lambda \;\Rightarrow\; z^* \;\Rightarrow\; \text{measurement closure.}$$

Two no-go theorems sharpen why no weaker hypothesis suffices: **Thm q12.dim2** (PROVEN) — a bare oriented 2-surface does not force the 4π spinor closure (its tangent frame is SO(2) = U(1), $\pi_1 = \mathbb{Z}$, and the spin structure is an $H^1(\Sigma; \mathbb{Z}/2)$-torsor choice); and **Thm q12.med** (PROVEN) — pure X–Y "mediation" admits at least six interface topologies, so it does not single out a codimension-1 boundary. What closes the gap is one statement in two equivalent forms (ZS-Q12V §15): **Z separates X from Y ⇔ the X–Y transfer is holographic ⇔ Z = ∂X** (Jordan–Brouwer: a separating hypersurface in the 3D X-bulk is codimension-1; Bekenstein: an area-saturating channel is the codim-1 boundary). This is the single irreducible postulate of the corpus.

ZS-Q16 v2.5 §22 then attempted to discharge exactly this residual by a spin-geometric realization and reported, honestly, that the route **decomposes** the embedding into three sub-conditions (Regge realization, normal-bundle reconstruction, oriented-frame closure) whose conjunction is **logically equivalent** to Z = ∂X — a clarification, not a reduction. Theorem Q16.C2 was therefore not promoted and remains DERIVED-CONDITIONAL.

Three consequences for ZS-S16. **(i)** The conditional of Theorem S16.1 is not an ad hoc gap specific to the lepton sector; it is the **same universal postulate** on which every PROVEN and DERIVED result in the corpus rests. **(ii)** It is in fact the very structure that **produces** the object this paper uses: the Wilson holonomy $\oint \omega = \boldsymbol{A}$ is the parallel transport on the codim-1 boundary $Z = \partial X$, so the sum rule's input and its conditional share one origin — the kick is not an extra assumption layered on the holonomy, it is the holonomy of the postulated boundary. **(iii)** Consequently the elevation of Theorem S16.1 and Lemma S16.5 to unconditional DERIVED is **pending the corpus-wide closure of $Z = \partial X$** (the one genuine lead being O-Q16.12: reconstructing the boundary normal bundle from the X-side record flow, an information→geometry bridge), and not pending anything internal to ZS-S16. In this precise sense the sum rule is as closed as any result in Z-Spin can be.

§4. arctan(A) as a Resummation Chart, Not a Physical Prediction

4.1 The chart-equivalence theorem

Theorem S16.2 (Chart Equivalence, PROVEN). For the locked value $\boldsymbol{A} = 35/437$ the Maclaurin expansion gives

$$\arctan(\boldsymbol{A}) = \boldsymbol{A} - \boldsymbol{A}^3/3 + \boldsymbol{A}^5/5 - \cdots,$$

so arctan($\boldsymbol{A}$) and $\boldsymbol{A}$ agree to first order with leading discrepancy $\boldsymbol{A}^3/3 = 1.71\times10^{-4}$ rad. Numerically arctan($\boldsymbol{A}$) = 4.5791° and $\boldsymbol{A}$ = 4.5889°, a difference of 0.0098° — four orders of magnitude below the best foreseeable resolution on δ_{CP} (Hyper-K best case ≈20°, [2]). The two surface forms are observationally identical.

The arctan form arises as the argument of a complex impedance written in the Cayley chart $Z_{eff} = 1 + i\boldsymbol{A}$, $\arg(Z_{eff}) = \arctan(\boldsymbol{A})$. We emphasize what this is and is not. It **is** a tidy coordinate in which the real part is the Z_2-even mode $\beta_0(Z) = 1$ (ZS-S1 §5.2) and the imaginary part is the seam-phase drift $\boldsymbol{A}$. It is **not** a derivation: a Hermitian Schur complement of the Z-sector yields a real correction (the β_0 shift 92→93 of ZS-S1 is real), and the Z_2-odd residual carried by the Wilson operator is real (0.205, ZS-F11/F12 Thm 12.3), not $i\boldsymbol{A}$. Hence the imaginary unit in $1 + i\boldsymbol{A}$ is a chart choice, and arctan($\boldsymbol{A}$) is the resummation it induces. The physical first-order content is the single real number $\boldsymbol{A}$.

4.2 When the exact arctan would be physical: the retarded Feshbach condition

There is exactly one way for the full arctan to acquire physical status. If the Z-sector is integrated out not by a static Hermitian Schur complement but by an **open-channel retarded Feshbach self-energy** [7],

$$\Sigma^R(E) = C_{XZ}\,[E - H_Z - \Sigma_Y^R(E)]^{-1}\,C_{ZX}, \quad \Sigma_Y^R = \Delta_Z - (i/2)\Gamma_Z\,,$$

then the effective coupling is genuinely non-Hermitian and its phase is

$$\varphi_Z = \arg(\Delta_Z - (i/2)\Gamma_Z) = -\arctan(\,\Gamma_Z / 2\Delta_Z\,).$$

The exact arctan($\boldsymbol{A}$) is then recovered if and only if the level-shift-to-width ratio satisfies $\Gamma_Z/(2\Delta_Z) = \boldsymbol{A}$. This is a sharp, well-posed open problem: derive the ratio from the Z-sector optical theorem under the $L_{XY} \equiv 0$ bottleneck. We register it as F-S16.7 [OPEN]. It is strictly stronger than the chart of §4.1 because it would promote arctan from convention to a consequence of a decay-width identity. The current corpus does not close it, so the exact arctan remains HYPOTHESIS-weak; the leading kick $\boldsymbol{A}$ survives regardless.

Version-conflict resolution (protocol 3.2). The chart equivalence closes the apparent ZS-Q5 (“$\delta_\varphi = \boldsymbol{A}$”, linear) vs ZS-S9 (“arctan($\boldsymbol{A}$)”) tension: $\arctan(\boldsymbol{A}) = \boldsymbol{A} - O(\boldsymbol{A}^3)$, so ZS-Q5's linear form is the leading term and ZS-S9's arctan is its chart. The two are compatible to 0.01°; no downstream prediction in ZS-S5, ZS-S10 shifts measurably.

§5. Corollary: the Conditional Jarlskog Suppression

Corollary S16.5.1 (Conditional Jarlskog Suppression, DERIVED-CONDITIONAL). An immediate consequence of Lemma S16.5 evaluated in the standard parametrization ($J = c_{12}s_{12}c_{23}s_{23}c_{13}^2\, s_{13} \sin\delta_{CP}$): at fixed mixing angles the leptonic Jarlskog invariant carries the parameter-free suppression

$$J / J_{bare} = \sin(\pm\pi/2 + s_Z\boldsymbol{A}) / \sin(\pm\pi/2) = \cos \boldsymbol{A} = 1 - \boldsymbol{A}^2/2 + O(\boldsymbol{A}^4).$$

Numerically $\cos \boldsymbol{A} = 0.996794$, a 0.321% suppression ($1 - \boldsymbol{A}^2/2 = 0.996793$ to $O(\boldsymbol{A}^4)$). Because J is rephasing-invariant, this ratio is convention-free — it removes the branch/ordering ambiguity that afflicts δ_{CP} itself. The arctan-chart variant gives $\cos(\arctan \boldsymbol{A}) = 1/\sqrt{1+\boldsymbol{A}^2} = 0.996808$, differing from $\cos \boldsymbol{A}$ only at $O(\boldsymbol{A}^4) \approx 1.4\times10^{-5}$ — unobservable. Status: DERIVED-CONDITIONAL (a corollary of Lemma S16.5; the rephasing-invariant Im[…] = (bare) cos A form is the lemma's own conclusion). The $1/\sqrt{1+\boldsymbol{A}^2}$ form is HYPOTHESIS-weak (it depends on the arctan chart).

5.1 The decoherence fork (a distinguishing prediction)

A second, physically distinct effect of the same Z-mediator must be separated from the phase kick. ZS-Q15 establishes the Z-mediator Lindblad coefficient $2\boldsymbol{A}/Q = \lambda_2$ (a real, dissipative decoherence rate). A dissipator **damps** coherence — it suppresses |J| — but it does **not** rotate the unitary phase, hence does not shift δ_{CP}. A dissipative J suppression would be of order $2\boldsymbol{A}/Q \approx 1.46\%$ — numerically distinct from, and larger than, the 0.32% phase-rotation suppression $\cos \boldsymbol{A}$. The two mechanisms are therefore a **fork**, separable in principle: a phase-rotation origin shifts δ_{CP} and suppresses J by 0.32%; a decoherence origin leaves δ_{CP} maximal and suppresses J by ≈1.5%. We register this as TESTABLE F-S16.6. (ZS-Q15's channel is the EM which-path sector; whether it acts on the PMNS flavor channel is itself OPEN.)

§6. The Contragredient Flavor-Orientation Theorem

v1.0 lumped “which sign” and “which lepton” into one statement. v1.1 separates them: the **relative orientation** (the sign s_Z in the contragredient frame) closes to DERIVED-CONDITIONAL, while the **absolute flavor label** (which physical lepton sits on the negative branch) is isolated as the lone residual OPEN.

Theorem S16.6 (Contragredient Flavor Orientation, DERIVED-CONDITIONAL). Given (i) the signed seam witness $\chi_Z = -1$ (ZS-Q14 Lem 5.1: a physical spinor Z-boundary in the nontrivial class of $H^2(SO(3), U(1)) \cong \mathbb{Z}_2$ forces the half-angle connection $G = \sigma_y/2$, sign-valued, parameter-free), and (ii) the contragredient pairing $V_{ZY} = (V_{XZ})^*$ (ZS-F4 §7B), the orientation of the off-diagonal spurion $B_{\mu\tau}$ of Lemma S16.5 is fixed to $s_Z = (\operatorname{sgn} \chi_Z)\cdot$(orientation of the contragredient pairing) $= -1$, selecting $\delta_{CP} = -\pi/2 - \boldsymbol{A} = 265.41°$ (inverted ordering).

Consistency check (corpus). ZS-S9 Cross-Check 5.1 independently routes the contragredient phase onto the down-type/charged-lepton effective Yukawa, $Y_e^{eff} = Y_e^{bare}\cdot e^{-i\boldsymbol{A}}$, entering the PMNS matrix asymmetrically and yielding the negative branch — the same sign Theorem S16.6 derives. The Z-Spin hardware protocol of ZS-Q14 (signed seam witness with negative controls) provides an

experimental/hardware corroboration of the sign: a measured $\chi_Z = +1$ would flip the orientation and refute the spin-½ lift basis (F-S16.8).

Residual OPEN (now isolated). The map from the contragredient seam frame to the **physical μ-vs-τ label** (tying the seam orientation to the charged-lepton mass ordering $m_\mu < m_\tau$) is the $U(1)_Z \leftrightarrow U(1)_Y$ action-level identification, which remains OPEN (ZS-S10 Gap G1; ZS-S9 Cor III strong sense). Theorem S16.6 fixes the relative orientation but not this absolute labelling; the opposite branch (+***A***) is carried by the CPT-conjugate (τ or anti-sector) once the label is fixed. This isolation — orientation DERIVED-CONDITIONAL, absolute label OPEN — is sharper than v1.0's combined S16.4.

§7. Observational Status and Falsification Gates

7.1 NuFIT 6.0 comparison

NuFIT 6.0 (data to Sept 2024 [1]) reports, for inverted ordering, δ_{CP} = 285° (+25/−28), with CP conservation excluded at >3.6σ; for normal ordering the fit is consistent with CP conservation within 1σ (best fit ≈97% near 177°). Our prediction δ_{CP} = 265.41° sits within ≈0.7σ of the IO best fit (lower 1σ edge ≈257°), but is **inapplicable to normal ordering**, where the data prefer near-CP-conservation — the opposite of the maximal ±π/2 seed. The global fit now mildly prefers normal ordering ($\Delta\chi^2 \approx$ 4.7–9.3 with Super-K atmospheric), so the sharp value is **IO-conditional**. We also note θ_{23} = 48.5° (second octant), already in mild tension with the μ–τ fixed point θ_{23} = 45° (an O(***A***) spurion stress).

Table 2. Prediction vs NuFIT 6.0 (degrees).

Quantity	Z-Spin (kick = A)	arctan chart	NuFIT 6.0	Status
δ_{CP} (IO, −branch)	265.41	265.42	285 (+25/−28)	≈0.7σ (IO)
offset from ±90°	4.589	4.579	< resolution	undiscriminable
J/J_{bare}	cos A = 0.99679	0.99681	n/a	DERIVED-COND
δ_{CP} (NO)	inapplicable	inapplicable	≈177	tension

7.2 DUNE / Hyper-Kamiokande kill timeline

DUNE determines the mass ordering at 5σ over the full δ_{CP} range and reaches 3σ (5σ) CP-violation sensitivity at ≈5 (10) years [3]; Hyper-Kamiokande measures δ_{CP} to ≈20° at maximal CP and ≈6° at CP conservation in 10 years [2], with a 5σ CPV discovery possible in <3 years for maximal CP and known ordering. The 4.59° offset is below the 20° maximal-CP resolution, so the offset is **not** individually detectable; but the conjunction "IO + near-270° + maximal CP" is a sharp, decidable kill gate this decade.

7.3 Falsification gates

Gate	Type	Condition (falsifies)	Status
F-S16.1	OBS	Ordering determined NORMAL at 5σ (DUNE): the IO −-branch sharp target is moot; lepton-sector mechanism loses its sharp prediction.	OPEN (NO-leaning)
F-S16.2	OBS, decisive	δ_CP measured CP-conserving (0 or π) at >5σ for the relevant ordering: kills the maximal ±π/2 seed (μ–τ reflection).	OPEN
F-S16.3	OBS	θ_{23} driven from 45° beyond the O(A) spurion bound: μ–τ reflection seed fails.	tension (48.5°)

Gate	Type	Condition (falsifies)	Status
F-S16.4	MATH	$\oint \omega \neq A$ on the primitive cell (contradicting ZS-F0/F-rotor): Thm S16.1 collapses.	PASS (safe)
F-S16.5	MATH	[CLOSED] A vs A/2 resolved: the PMNS reads the rephasing-invariant relative phase arg(V_XZ)−arg(V_ZY)=A; A/2 is the non-invariant single-path phase (§3.4).	PASS
F-S16.6	OBS, fork	J suppression measured ≈ 2A/Q (≈1.5%) with δ_CP maximal: decoherence origin, not phase rotation.	OPEN (far-future)
F-S16.7	MATH	Retarded Feshbach gives Γ_Z/(2Δ_Z) ≠ A: exact arctan excluded (leading kick A survives).	OPEN
F-S16.8	HW	χ_Z = +1 on Z-Spin hardware (ZS-Q14 F-Q14.1): spinor-lift basis of s_Z fails.	OPEN

§8. Zero Free Parameter & Anti-Numerology Audit

8.1 Parameter audit

Every quantity in the sum rule is locked or imported: $\boldsymbol{A}$ = 35/437 (LOCKED, ZS-F2); Q = 11, dim Z = 2 (PROVEN, ZS-F5); $\delta_{\mu\tau} = \pm\pi/2$ (IMPORTED, μ–τ reflection); the kick magnitude $\boldsymbol{A}$ is the DERIVED Wilson holonomy (no continuous coefficient); the sign $s_Z \in \{-1,+1\}$ is a discrete two-valued choice fixed by $\chi_Z = -1$. No continuous parameter is introduced. **Free parameters: zero.**

8.2 Anti-numerology: why A and not arctan A, A/2, or c·A

The decisive anti-numerology improvement over the original arctan proposal is that the magnitude is no longer chosen from a continuum of "forms ≈ $\boldsymbol{A}$" but is **fixed by a closed-form holonomy** $\oint \omega = \boldsymbol{A}$ (DERIVED). A pre-registered Monte Carlo over the discrete candidate set {$\boldsymbol{A}$, arctan $\boldsymbol{A}$, $\boldsymbol{A}$/2, 2$\boldsymbol{A}$, c·$\boldsymbol{A}$ for random c} confirms that only the single-cell holonomy coefficient c = 1 is forced by the Gauss–Bonnet identity; the remaining members are either chart resummations of the same leading term (arctan $\boldsymbol{A}$, $\boldsymbol{A} - \boldsymbol{A}^3/3$) and hence physically identical to $O(\boldsymbol{A}^3)$, or double-cover variants ($\boldsymbol{A}$/2, registered as F-S16.5). The corpus precedent ZS-A14 reports the SU(2)-cell anti-numerology at MC hit 0.62–0.64% (< 5%), consistent with this audit. Unlike a symbolic-assignment test (which the corpus showed cannot beat its own granularity, ZS-S9 App. B), the present claim is a numerical magnitude with a derivation chain, so it is testable in the proper sense.

8.3 Methodological contribution: the chart-vs-physics discipline

The external, transferable lesson is a discipline applicable to any "small geometric correction" model: **distinguish a first-order physical prediction from a resummation chart**. When a small correction ε admits multiple surface forms (ε, arctan ε, sin ε, $\varepsilon/\sqrt{(1+\varepsilon^2)}$, …) that coincide to $O(\varepsilon^3)$, the only defensible physical claim is the leading coefficient, fixed by a closed-form invariant; the choice of transcendental wrapper is a coordinate convention and must be labelled NON-CLAIM unless an independent mechanism (here, a retarded Feshbach width ratio, §4.2) promotes it. This principle generalizes beyond Z-Spin to any beyond-Standard-Model flavor sum rule.

§9. Cross-Paper Dependency and Version-Conflict Check

Table 3. Dependency ledger (protocol 3.2).

Upstream	Role in ZS-S16	Status used
ZS-F0/F-rotor §5.8	$\oint \omega$ = A primitive-cell Wilson holonomy (the anchor of Thm S16.1)	**DERIVED**
ZS-F2	A = 35/437 geometric impedance	**LOCKED**
ZS-F4 §7, §7B	half-angle V_XZ, contragredient V_ZY = (V_XZ)*	**DERIVED-COND**
ZS-F5	(Z,X,Y)=(2,3,6), Q=11, Z_2 seam W^2=I	**PROVEN**
ZS-M3 Thm 5.1	j = 1/2 uniqueness, D(2π) = −I (half-angle)	**PROVEN**
ZS-Q5 §4	δ_φ = A impedance phase (reinterpreted as leading term)	**DERIVED-COND**
ZS-Q14 Lem 5.1	χ_Z = −1 spin-½ lift selection (sign s_Z)	**DERIVED-COND**
ZS-Q15	Lindblad 2A/Q decoherence fork (§5.1)	**DERIVED**
ZS-S5 §3.2, §8	A-controlled Z_2 breaking; A≠0 ⇒ J_CP≠0; closes OP-(ii)	**DERIVED**
ZS-S5 App. B	bounded spurion ε = O(A) (existence + magnitude of B_μτ, Lemma S16.5)	**DERIVED**
ZS-S9 Cor III	branch existence (weak DERIVED / strong OPEN)	**DERIVED/OPEN**
ZS-S10	U(1)_Z ↔ U(1)_Y identification (Gap G1)	**DERIVED**

Version-conflict result: promoting ***A*** over arctan(***A***) changes no downstream numerical prediction by more than 0.01° (δ_{CP}) or 1.4×10^{-5} (J), so all cross-citing results (ZS-S5 leptogenesis, ZS-S10 gauge bridge) remain safe. The reinterpretation is strictly additive (no retraction).

§10. Non-Claims

NC-S16.1. We do NOT claim arctan(***A***) is the exact physical CP phase; it is a resummation chart (Thm S16.2). The physical first-order content is ***A***.

NC-S16.2. We do NOT claim δ_{CP} = 265.41° is a currently-closed prediction; it is IO-conditional and registered as a kill gate.

NC-S16.3. We do NOT claim the leptonic Jarlskog is measured; cos ***A*** is a DERIVED-CONDITIONAL theoretical suppression, not VERIFIED.

NC-S16.4. We do NOT derive which charged lepton carries the negative branch (ZS-S9 Cor III strong sense remains OPEN); only the existence and the sign of the kick relative to the seam are fixed.

NC-S16.5. [Resolved in v1.1] The full-angle/half-angle (***A*** vs ***A***/2) question is no longer open: §3.4 shows the PMNS reads the rephasing-invariant relative contragredient phase = ***A***, so ***A***/2 (the non-invariant single-path phase) is a closed rebuttal.

NC-S16.6. We do NOT fix the absolute μ-vs-τ flavor label of the negative branch; Theorem S16.6 fixes only the relative orientation, the absolute label remaining OPEN (ZS-S10 Gap G1).

§11. Conclusion

We have reconstructed the Z-Spin leptonic CP prediction on a footing that survives external scrutiny. The weak link was always the transcendental wrapper arctan(***A***); removing it and promoting the locked, DERIVED Wilson holonomy ***A*** to the primitive first-order CP phase kick yields the sum rule $\delta_{CP} = \pm\pi/2 + s_Z \boldsymbol{A} + O(\boldsymbol{A}^3)$, with a rephasing-invariant Jarlskog suppression $\cos \boldsymbol{A} = 1 - \boldsymbol{A}^2/2$ and a branch sign $s_Z = -1$ fixed (DERIVED-CONDITIONAL) by the spin-½ boundary lift. The arctan form is demoted to a Cayley chart, and the exact-arctan question is sharpened to a single open width-ratio identity $\Gamma_Z/(2\Delta_Z) = \boldsymbol{A}$ (F-S16.7). The sharp value $\delta_{CP} = 265.41°$ is honestly registered as an IO-conditional kill gate against a now-NO-leaning global fit, decidable by DUNE and Hyper-Kamiokande this decade. Beyond Z-Spin, the work contributes a transferable discipline — separating a physical first-order prediction from a resummation chart — and a concrete open bridge between non-Hermitian self-energy phases and a geometric constant. Zero free parameters throughout.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools for cross-paper integration, symbolic verification, and manuscript drafting under the editorial direction of the author, who assumes full responsibility for all scientific content. The verification suite zs_s16_verify_v1_2.py (32/32 PASS, categories A–H plus the Lemma S16.5 non-absorbability and cos A scaling checks; NumPy/mpmath) reproduces all numerical claims, including the corrected degree conversions of §4.1/Appendix A; the anti-numerology Monte Carlo and the Table 4 checks are included. No new theoretical constants; ($\boldsymbol{A}$, Q, dim Z) = (35/437, 11, 2) LOCKED.

Appendix A. Trigonometric Identities of the Sum Rule

A.1 Chart equivalence: $\arctan(x) = x - x^3/3 + x^5/5 - \cdots$; at $x = \boldsymbol{A} = 0.0800915$, $\arctan(\boldsymbol{A}) = 0.0799208$ rad, $\boldsymbol{A} - \boldsymbol{A}^3/3 = 0.0799203$ (agreement to 6 digits).

A.2 Jarlskog: $\sin(\pi/2 + \varphi) = \cos\varphi$ for any φ, hence $J/J_{bare} = \cos \boldsymbol{A}$ with kick $\boldsymbol{A}$, and $= \cos(\arctan \boldsymbol{A}) = 1/\sqrt{1+\boldsymbol{A}^2}$ with kick $\arctan \boldsymbol{A}$. The two differ at $O(\boldsymbol{A}^4)$: $\cos \boldsymbol{A} = 0.9967944$, $1/\sqrt{1+\boldsymbol{A}^2} = 0.9968080$, $\Delta = 1.36\times10^{-5}$.

A.3 Degrees: $\boldsymbol{A} = 4.58891°$, $\arctan(\boldsymbol{A}) = 4.57913°$, $\Delta = 0.00978°$; $\boldsymbol{A}/2 = 2.29443°$.

Appendix B. The Retarded Feshbach Open Problem

Static Hermitian elimination of the Z-sector (Schur complement, ZS-S1 §5.3) gives a real correction; no CP phase. An *open-channel* elimination uses the retarded self-energy $\Sigma^R(E)$ with imaginary part $-(i/2)\Gamma_Z$ (the optical-theorem width). Its argument is $-\arctan(\Gamma_Z/2\Delta_Z)$. Recovering the exact $\arctan(\boldsymbol{A})$ requires $\Gamma_Z/(2\Delta_Z) = \boldsymbol{A}$. Open problem (F-S16.7): derive this ratio from the Z-sector optical theorem under the $L_{XY} \equiv 0$ bottleneck and the Y/Q spectral density. A solution would upgrade $\arctan(\boldsymbol{A})$ from chart (NON-CLAIM) to a consequence of a decay-width identity (DERIVED-CONDITIONAL); failure leaves the leading kick $\boldsymbol{A}$ intact.

Appendix C. Numerical Verification Table

Table 4. Verification suite (32/32 PASS), representative checks.

Cat.	Check	Value	Result
A1	A = 35/437 (lowest terms, gcd=1)	0.0800915	PASS
A2	$\oint \omega = A$ (ZS-F0/F-rotor §5.8)	= A	PASS
B1	$\arctan(A) - (A - A^3/3)$	5×10^{-7}	PASS
B2	$\sin(\pi/2 + A) - \cos A$	0 (exact)	PASS

Cat.	Check	Value	Result
B3	$\cos(\arctan A) - 1/\sqrt{1+A^2}$	0 (exact)	PASS
C1	A in degrees	4.58891°	PASS
C2	arctan(A) in degrees	4.57913°	PASS
C3	$\lvert A - \arctan A\rvert$ in degrees	0.00978°	PASS
C4	J/J_bare = cos A (suppression)	0.99679 (0.321%)	PASS
C5	$\cos A - 1/\sqrt{1+A^2}$	1.36×10^{-5}	PASS
D1	$\delta_{CP} = -\pi/2 - A$ (IO −branch)	265.411°	PASS
D2	Lemma S16.5 (P1): $\mathrm{Tr}(B_{\mu\tau} \cdot D)=0$ for all diagonal D	0 (off-diag)	PASS
D3	A vs A/2: $\arg(V_{XZ})-\arg(V_{ZY})=2(A/2)=A$ (rephasing-inv.)	= A	PASS
E1	pull vs NuFIT 6.0 IO bfp 285°	≈0.7σ	PASS
F1	decoherence fork 2A/Q	1.46%	PASS
G1	anti-numerology c=1 forced (MC, ZS-A14 0.62%)	< 5%	PASS
H1	8 falsification gates registered (F-S16.5 now CLOSED/PASS)	F-S16.1–8	PASS

References

[1] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, T. Schwetz, “NuFit-6.0: updated global analysis of three-flavor neutrino oscillations,” J. High Energy Phys. 12 (2024) 216; arXiv:2410.05380.

[2] Hyper-Kamiokande Collaboration, “Sensitivity of the Hyper-Kamiokande experiment to neutrino oscillation parameters using accelerator neutrinos,” Eur. Phys. J. C 85 (2025); arXiv:2505.15019.

[3] DUNE Collaboration, B. Abi et al., “Long-baseline neutrino oscillation physics potential of the DUNE experiment,” Eur. Phys. J. C 80 (2020) 978; arXiv:2006.16043.

[4] P. F. Harrison and W. G. Scott, “μ–τ reflection symmetry in lepton mixing and neutrino oscillations,” Phys. Lett. B 547 (2002) 219; arXiv:hep-ph/0210197.

[5] C. Jarlskog, “Commutator of the quark mass matrices in the standard electroweak model and a measure of maximal CP nonconservation,” Phys. Rev. Lett. 55 (1985) 1039.

[6] Particle Data Group, R. L. Workman et al., “Review of Particle Physics,” Prog. Theor. Exp. Phys. 2024, 083C01 (2024).

[7] H. Feshbach, “Unified theory of nuclear reactions,” Ann. Phys. (N.Y.) 5 (1958) 357; ibid. 19 (1962) 287.

[8] K. Kang, “ZS-F0 / F-rotor: Time–Rotation Co-Definition and the Wilson-loop holonomy $\oint \omega = A$,” Z-Spin Cosmology Collaboration (2026).

[9] K. Kang, “ZS-F2: The Geometric Impedance A = 35/437,” Z-Spin Cosmology Collaboration (2026).

[10] K. Kang, “ZS-F4 §7B: Contragredient Half-Holonomy $V_{ZY} = (V_{XZ})^*$,” Z-Spin Cosmology Collaboration (2026).

[11] K. Kang, “ZS-M3: Spin-½ Uniqueness of the Z-Sector Intertwiner,” Z-Spin Cosmology Collaboration (2026).

[12] K. Kang, “ZS-Q5: Z-Mediated Geometric Impedance Phase $\delta_\varphi = A$,” Z-Spin Cosmology Collaboration (2026).

[13] K. Kang, “ZS-Q14: Canonical Kraus-Lift Selection by a Spin-½ Boundary,” v1.4, Z-Spin Cosmology Collaboration (2026).

[14] K. Kang, “ZS-Q15: Electromagnetic Which-Path Decoherence and the Block–Fiedler Coupling $\kappa^2 = A/Q$,” Z-Spin Cosmology Collaboration (2026).

[15] K. Kang, “ZS-S5: Resonant Leptogenesis from the Neutrino Sector,” Z-Spin Cosmology Collaboration (2026).

[16] K. Kang, “ZS-S9: The Electron Synthesis Theorem (Cor III, Cross-Check 5.1),” Z-Spin Cosmology Collaboration (2026).

[17] K. Kang, “ZS-S10: Gauge Bridge via the Stückelberg–Corollary IV Mechanism,” Z-Spin Cosmology Collaboration (2026).

Version History

v1.2 (June 2026): Positioning + errata release (strictly additive; no claim retracted). (a) NEW Sec. 3.5 Bedrock Positioning: the lone conditional of Theorem S16.1 and Lemma S16.5, the embedding Z = dX, is identified with the single irreducible postulate of the entire Z-Spin corpus via the ZS-Q12V Sec.13-16 Bedrock cascade (Z=dX => dim Z=2 => SO(3) => 4pi => i-tetration => z*); the two no-go theorems (q12.dim2, q12.med) and the Jordan-Brouwer/Bekenstein closure are summarised, and ZS-Q16 v2.5 Sec.22 is cited as the honest demonstration that the residual decomposes but cannot be reduced further. Consequence: the sum rule is as closed as any Z-Spin result; unconditional DERIVED is pending only the corpus-wide closure of Z=dX (O-Q16.12), not anything internal to ZS-S16. (b) Erratum: degree conversions corrected to A = 4.58891 deg, arctan(A) = 4.57913 deg, |A - arctan A| = 0.00978 deg (Sec.4.1, Appendix A.3, Table 4); 5th-significant-figure rounding slips in v1.0-v1.1, immaterial to every conclusion and falsification gate. The high-precision leading term is cos A - 1/sqrt(1+A^2) = -A^4/3 + O(A^6). (c) Verification suite renamed zs_s16_verify_v1_2.py; the suite computes the corrected degree values, so it already matched the v1.2 text. 32/32 PASS, zero free parameters, (A, Q, dim Z) = (35/437, 11, 2) LOCKED.

v1.1 (June 2026): Strengthening release (four reviewer patches). (1) NEW Lemma S16.5 (Rephasing-Invariant PMNS Spurion): the breaking is a real, off-diagonal, mu-tau-seam-odd, CP-odd spurion Delta_Z = i s_Z A B_mutau, proven orthogonal to the charged-lepton rephasing algebra (hence non-absorbable, not a diagonal/Majorana rephasing) and yielding Im[...] = (bare) cos A + O(A^3); this upgrades Theorem S16.1 from an embedding postulate to an invariant derivation and closes the registered corpus open problem ZS-S5 OP-(ii). (2) The full-angle/half-angle question is resolved (Sec. 3.4, Patch 2): the PMNS reads the rephasing-invariant relative contragredient phase arg(V_XZ)-arg(V_ZY)=A, so A is selected and A/2 is a closed rebuttal; gate F-S16.5 moves OPEN to PASS. (3) Theorem S16.3 (Jarlskog) demoted to Corollary S16.5.1 of Lemma S16.5 (Patch 4). (4) Theorem S16.4 (Branch) replaced by Theorem S16.6 (Contragredient Flavor Orientation, Patch 3), separating the relative orientation s_Z=-1 (DERIVED-CONDITIONAL) from the absolute mu-vs-tau label (OPEN, ZS-S10 Gap G1). Verification 28/28 to 32/32 PASS, zero free parameters, (A, Q, dim Z) = (35/437, 11, 2) LOCKED.

v1.0 (June 2026): Initial public release. Consolidated from internal Z-Spin Collaboration deep-exploration notes (four convergent cycles) up to The Book of Z-Spin Cosmology v7.0. Core reconstruction: the leptonic CP prediction recast from delta_CP = +/-pi/2 + arctan(A) to the Wilson-holonomy sum rule delta_CP = +/-pi/2 + s_Z*A + O(A^3); arctan(A) demoted to a Cayley resummation chart (Thm S16.2); exact-arctan reduced to the open width-ratio identity Gamma_Z/(2 Delta_Z) = A (F-S16.7). 28/28 PASS.